Sheriff case SHF-20261021-477d51: crypto PR #2343 (IC planner and its validation)

Target. Everything aburan28/crypto PR #2343 merged to `main` (merge commit `b4f056101`, 17 commits of producer session `session_01R35qGy…`):

- the `ic plan` / `ic bench --auto` planner (`src/cryptanalysis/ic_framework/planner.rs`, `src/bin/ic/{plan,bench}.rs`);
- the validation study `research/ic_planner_validation_20261020` (six ecbench sessions);
- the 12 restored `plan.json` files under `research/f4_pricing_jump_20261018`;
- the "Planner validation" leaderboard section, scoreboard rows and the `ec-cold` progress point 14.39;
- the `ic-autolab.yml` RUNPATH symlink and zstd change.

The tier is deep.

Audit record only. This case changes no research state. It repairs nothing in crypto, and no file in crypto was edited.

Independence. This session (`session_01WAo1Hy…`) appears in none of the audited commits (`checks/session-checks.txt`).

Prior (first, as written before reading; `prior.md`)

I expected the following:

1. A protocol or amendment not strictly before its sessions.
2. Wording that generalises six toy-size cells.
3. `rho.negation` correctly matched, but with unsourced rho tables in the planner.
4. At least one planner citation that does not support its rule, and a fitted or recalled $\kappa = 0.6$.
5. The restored `plan.json` files legitimate but weakly evidenced.
6. A leaderboard or progress point that reads as a gain.
7. The workflow keeping its gate.

How the prior fared.

- (1) was *not* borne out.
- (2), (4) and (6) were borne out.
- (3) was half right: the tables trace to committed files, but carry `Measured` labels they have not earned.
- (5) came out better than expected: the restored bytes equal the `plan_sha256` sealed two days earlier.
- (7) was borne out.

Instrument limits, stated first

This session's permission policy **denied building or running any code from crypto**. That covers `cargo build --release --bin ecbench`, `ecbench` itself, and the producer's scripts. So:

- **No `ecbench verify` replay was run.** The request asked for real replays on at least two sessions. That check is `not_covered`. The command is under "Not covered" below.
- **`ecbench compare`, `table` and `band` were not re-run.** The stats reader ported their methods to its own Python from a read-only reading of the source. The port matches all 1,536 producer values to 1e-9, all 396 values in `summary.json`, and all 156 values decoded from the figures.
- **The sweep ran text scans only (`--no-checkers`).** Its crypto-side checkers execute crypto code. The canary passed (15/15 caught, no false positives).

The sweep's other leads were dismissed:

- The two zk leads (`src/zk/groth16.rs`, `src/zk/plonk.rs`) and the four `tools/lean/selftest` leads are in files this PR did not touch.
- The `prose.strong_claim` leads in `benchmarks/README.md` are all older rows, except line 12, which is this PR's row and is covered by C5.
- The 200 `outside_sparse` rows are paths outside this PR.

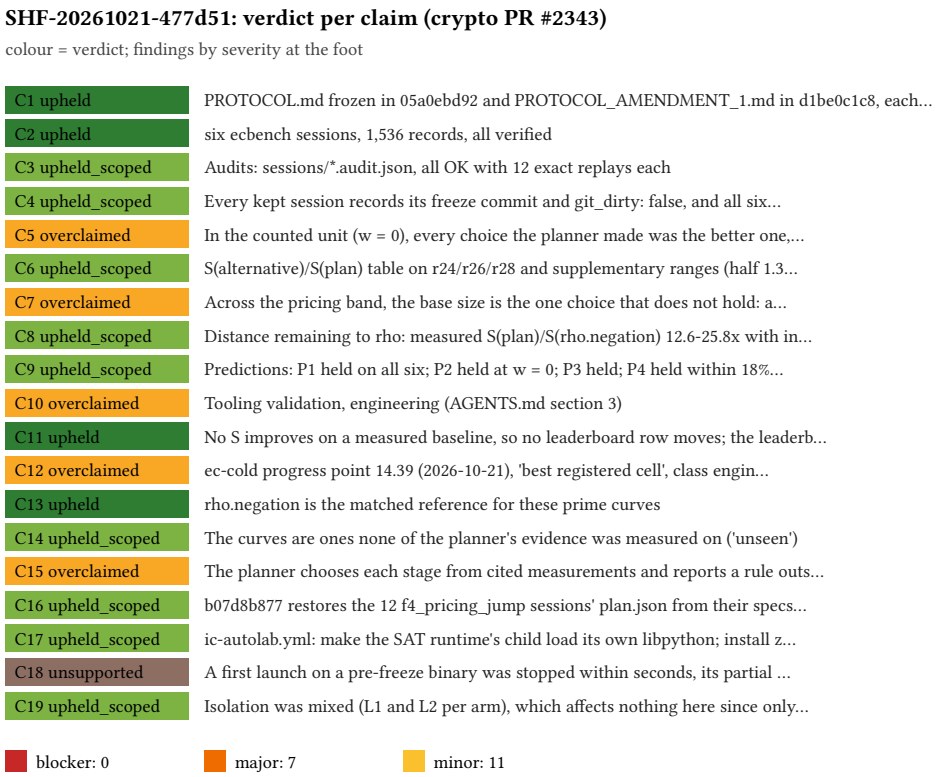


Figure 1: Verdict per claim (verdicts.svg, from findings.yaml).

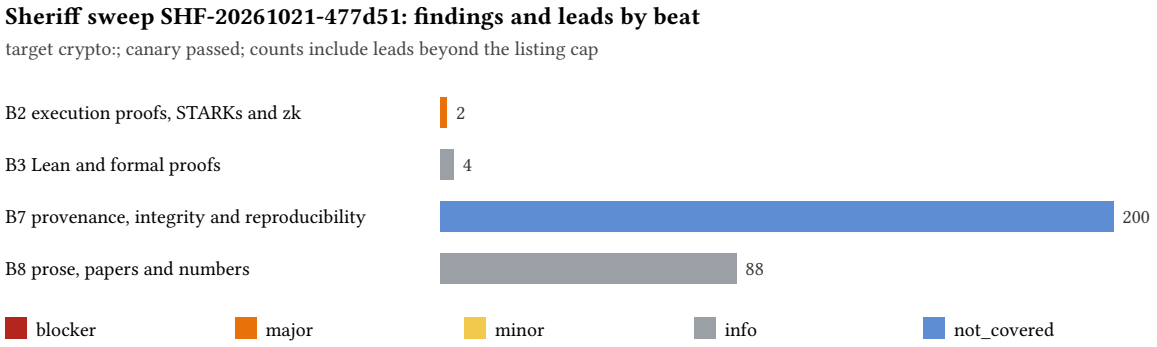


Figure 2: Deterministic sweep (sweep.svg): findings and leads by beat; canary passed; crypto-side checkers not run.

Verdicts

claim	producer's words (short)	verdict
C1	protocol frozen in 05a0ebd92, amendment in d1be0c1c8, each before its sessions; planner unchanged	**upheld**
C2	six sessions, 1,536 records, all verified	**upheld**
C3	audits OK, 12 exact replays each	upheld_scoped
C4	all six ran the same binary d0527101…7771, clean tree	upheld_scoped
C5	"in the counted unit, every choice the planner made was the better one, on all six curves"	**overclaimed**
C6	S(alt)/S(plan) table and supplementary ranges	upheld_scoped
C7	doubled base 2–8% / 4–9% cheaper at w = 1; "not separated"	**overclaimed**
C8	distance to rho 12.6–25.8×, predictions inside intervals, within 18%	upheld_scoped
C9	P1 held on all six; P1–P4 held; P5 failed for double base	upheld_scoped
C10	class "engineering" (AGENTS.md §3)	**overclaimed**
C11	no leaderboard row moves; rows equal the frozen files	**upheld**

C12	`ec-cold` progress point 14.39, "best registered cell", engineering	**overclaimed**
C13	`rho.negation` is the matched reference for these curves	**upheld**
C14	the curves are unseen by the planner's evidence	upheld_scoped
C15	planner rules evidence-backed; outside measured sizes "extrapolated, never known"	**overclaimed**
C16	12 `plan.json` restored as a deterministic expansion, hashing to `plan_sha256`	upheld_scoped
C17	ic-autolab change: child loads its own libpython; zstd; gate intact	upheld_scoped
C18	first launch "stopped within seconds … deleted unread"	unsupported
C19	mixed L1/L2 isolation "affects nothing here"	upheld_scoped

Totals:

- **Claims:** 4 upheld, 9 upheld_scoped, 5 overclaimed, 1 unsupported, 0 broken.
- **Findings:** 0 blockers, 7 major, 11 minor.

Blockers

None. No answer is wrong, no seal was modified, no protocol was edited after data, and no record was dropped from the analysis.

Major findings

F1 (C5): the headline ranks at one weight, and generalises past what was tested.

- "Every choice the planner made was the better one" ranks at w = 0, which §5a forbids ("rank only where the conclusion holds at every weight"). Against the doubled base, the order reverses at w = 1 on all six curves.
- "Every choice" also covers more than was tested:
  - the base was tried only at ×0.5 and ×2, and the red team's proves-too-much control shows any N in about [0.78N, 1.41N] passes that test;
  - the oracle was tried only against `rootfind` (known 31–64× worse), never against m = 3 or the unblocked builder.
- The README says no IC wall time is used. It does not say that the admitted wall comparison (median 0.68–0.76 on all six curves, outside A/A noise) also favours the doubled base. I confirmed the six medians in `sessions/*/comparisons/plan__double-base.json`.
- **Fix:** rescope the headline to the five tested alternatives at w = 0, and call the base size unresolved across the band. Add the wall median as a practicality note. Run the cheapest falsification on the registered specs: base arms at 1.125N and 1.25N, plus m = 3 and unblocked-builder arms.

F2 (C7, C9): the decision rule applied is not the frozen one.

- `scripts/summary.jq:20-22` checks an interval only at w = 0, and point means at w = 0.1 and 1. It can never return "refuted" above w = 0. I confirmed this by reading the script. `PROTOCOL.md:68-71` requires the interval at every band weight. The deviation is not listed under Deviations.
- Per-weight intervals make the double-base label depend on the method:
  - the stats reader's two-stage interval contains 1 on all six curves;
  - the red team's one-stage paired interval lies below 1 on r28 [0.880, 0.991] and c30 [0.914, 0.999].
- The per-curve detectable effect (about 10%) is larger than the 2–9% effect, so "not separated" reflects a lack of power, not equivalence.
- **Fix:** record the deviation, compute per-weight intervals with one stated method, report the base size as "unresolved, underpowered", and pre-register the method and N for the follow-up.

F3 (C10, C12): misclassed as "engineering".

- Crypto `AGENTS.md` §3 defines engineering as "`S` fell … an improvement to the baseline: a success". The study itself says no `S` fell.
- The label appears in the README, the REPORT, the leaderboard section, the scoreboard and the progress chip.
- **Fix:** relabel it as a validation or diagnostic with no class gain.

F4 (C12): the progress point reads as frontier progress.

- The `ec-cold` point 14.39 is r24's w = 0 S(plan)/S(rho) = 14.3905 [11.63, 18.45]. It traces correctly.
- But it is the *best* of three registered cells, and a quotient of two lower-bound arms.
- It is placed in the series defined as "the repository's own definition of faster than rho", with an engineering chip and no lower-bound or diagnostic label.
- It moves no frontier or best value. The series' last point, 4.669 (rule v8), stays lower.
- **Fix:** move it to the bounded-counts series, or label it, and use the declared primary cell or all three, not the best.

F5 (C3, C6): no independent replay, and none at all for the five baseline arms.

- `ecbench verify --replay N` samples evenly spaced records (`audit.rs:489-499`). With 8 arms in alternate order, `--replay 12` landed only on `rho-neg` and `plan-aa`. The counts of the five baseline arms, the denominators of every ratio, were never replayed.
- CI runs `--replay 0`. Every replay ran on the producer's own host class.
- The sheriff could not replay (execution denied).
- **Fix:** run `--replay 192` per session on another host class, and stratify the replay sampler by arm.

F6 (C15): the planner's honesty contract is not implemented.

- `planner.rs:9-11` promises that a rule applied outside its measured sizes is reported as extrapolated, never as known. The code does not do this:
  - the prime size rule is `Support::Measured` over 23.4–32 bits, yet Table B's "family optimum" is the rule's own output, and the 32-bit grid starts at 1,024, above the rule's 813;
  - the oracle, target and linalg choices are `Measured` at every size;
  - `planner.rs:480` says "m = 2 beats m = 3 and root finding by 4–60×", while Table B has m = 3 at 1.05–3.2× and rootfind at 31–64×.
- **Fix:** honest support labels, and reason strings that quote the cited numbers.

F7 (C15): the wrong rho reference for j = 0 and 1728.

- For j = 0 and j = 1728 the planner names `rho.negation` as the matched rho (`planner.rs:643-648`).
- ecbench ships `rho.cubic` and `rho.quartic`, and the planner's own cited GLV note gives the matched folded rho $\sqrt{3}$ / $\sqrt{2}$ fewer steps. So the planner misstates the gap there.
- The campaign does not exercise this path.
- **Fix:** name the folded rho as the reference, or mark the outlook unmeasured for those curves.

Minor findings

F8: number slips.

- The r24 rho-interval upper bound, 18.4487, is printed as **18.5**. It should be 18.4: `README.md:77`, `REPORT.typ:55`, both PDFs. I confirmed the value in `analysis/summary.json:59`.
- "five native counters unpriced" (`README.md:55`) is wrong for the rootfind arm, which has six.
- "P1–P4 held on all six" (`REPORT.typ:70`) omits "at w = 0".

F9: the A/A control is degenerate. `plan-aa` shares `plan`'s seeds, so the pairs are identical and 1.000 [1, 1] only checks determinism. An A/A with different seeds, built from the same data, gives about [0.92, 1.09].

F10: the dropped first launch left no trace. It was "deleted unread", and no metadata survives to verify that (C18 is unsupported).

F11: the amendment came after the original data. It was frozen after the c-session data were committed, and it re-designated which sessions are primary. It is disclosed, only the curves changed, and the results agree.

F12: L2 isolation was not met for whole arms. This is disclosed, and the counts are unaffected (identical on 45 mixed-level pairs).

F13: the restoration changed sealed directories. The 12 `plan.json` files added to sealed f4 sessions are byte-identical to the sealed `plan_sha256`. But the restoration is not documented beside those sessions, they now pass `verify` only because of it, and their `git_commit` was already null.

F14: the binary carries no build commit. It records `git_commit_source runtime_worktree`, so the binary hash is not bound to a build.

F15: two binding weaknesses.

- Target isolation is by function signature, not by process: the measured child holds the planted k in memory, though no solver path reads it.
- The `prime_search` specs pin their curve only together with a rand-crate version.

F16: unlabelled outlooks and an unmeasured constant.

- The rho outlooks are not labelled as w = 0 lower-bound quotients.
- The prime end points were measured at N = 48 and 1,024, not at the rule's 41 and 813.
- The Koblitz outlook mixes units.
- $\kappa = 0.6$ is a hand fit to eight unkept picks. Six small curves pin it to [0.561, 0.629); it runs 25–33% high at n37/n41, and the source model gives 0.794.
- No `ic plan` output is committed for the six curves.

F17: a pre-existing gap in the SAT interpreter gate. It hashes the `python3` launcher, not `libpython`. zstd is installed without a version assertion.

F18: context outside the PR. `6a3605965` added this row to `benchmarks/PRE_GATE.json`, a file marked "may only shrink".

Per-beat results

B1 (claims). 19 claims were inventoried, including claims by omission:

- the wall comparison left out of the README (F1);
- the A/A presented without saying it is a determinism check (F9);
- the w = 0 scope missing from P1–P4 (F8).

B4 (certificates). Pass. Checked with independent arithmetic:

- [k]G = Q and recovered = planted on 1,536/1,536 records;
- k re-derived from `uniform_scalar_sha256_v1`;
- p and r prime, [r]G = O, and Hasse forcing #E = r;
- a and b nonzero, j ≠ 0, 1728;
- r24/r26/r28 equal their registry entries;
- the c-curves solved from their points and matched to their ICV1 hashes. Target independence holds on the code path (F15 minor).

B5 (cost model). Fail on F1, F3, F4, F6 and F7.

- Pass on matched rho for the six curves: `automorphisms_used` = 2, the same `ecbench.gae` unit and S = total/√r on both arms.
- Pass on phases: every phase present, setup charged cold per target.
- Pass on the band: recomputed to 1e-11.
- The gap to rho is reported as **distance remaining, not pass/fail**, in the README ("These are the distances remaining, not a pass/fail gate"), the REPORT and the leaderboard. The one framing issue is that it is not labelled as a w = 0 lower-bound quotient (F16): it is 15.6–30.7× at w = 1.
- The planner's predictions (11.46/19.33/24.55; 12.23/14.45/20.08) and base sizes (113/300/469; 128/175/323) re-derive exactly from its formulas and tables.
- `PRIME_RHO_POINTS` (19.10 ⟶ 6.6671; 32.0 ⟶ 32.955) and the binary and Koblitz points trace to committed files. The interpolation is log-linear in log2 r, and the label "interpolated" is correct for the campaign's sizes.

B6 (statistics and preregistration). Freeze order passes.

- **Ancestry:** each protocol commit is the direct parent of its data commit.
- **Run times:** the sessions ran after their freeze (c 17:00–17:16Z after 16:55:46; r 17:40–17:46Z after 17:35:49).
- **Run commit:** each session ran at its freeze commit as worktree HEAD, clean.
- **No later edits:** neither protocol file was edited after its freeze commit (`git log --follow`, at `b4f056101` and at `origin/main`).
- **Same code:** `src/`, `Cargo.toml` and `Cargo.lock` are identical from `a29bead23` through `b4f056101`.
- **N:** N planned = N written = 256 per session, all verified, no failure rows, warm-ups excluded.
- **Fail:** the decision rule as implemented (F2), and the best-of-3 progress point (F4).
- **Minor:** F9–F12.

B7 (provenance).

- Seals pass: records, spec, plan and host hashes and session ids recompute for all 18 sessions. Every campaign file has exactly one commit, its add, so nothing was resealed.
- The `plan.json` reconstruction is a deterministic expansion (C16).
- Replays are `not_covered` (F5).
- The binary hash is `not_covered` (F14).
- Workflow: the gate code and records are unchanged and still fail closed; a dangling link fails the step (`checks/workflow-ic-autolab.md`).

B8 (prose).

- 207/209 prose numbers match (F8).
- `REPORT.pdf` (created 03:13:03Z) post-dates the last `.typ` and number change (commit 03:13:25Z), and its text equals `REPORT.typ` (`checks/REPORT.pdf.txt`).
- All copies under `benchmarks/20261020-ic-planner-validation/` are byte-identical to the study's files.
- The leaderboard rows equal `summary.json` (C11).

Not applicable. B2 (no execution-proof language is used) and B3 (no Lean in this PR).

Not covered, and the command that would cover it

- **Replays:** `ecbench verify --dir research/ic_planner_validation_20261020/sessions/<s> --replay 192` for each of the six sessions, on another host class. This needs `cargo build --release --bin ecbench`, which was denied here. Use 192, not 24: with this sampler, 24 also lands only on `rho-neg` and `plan-aa`.
- **ecbench re-runs:** `compare`, `table` and `band` with the producer's binary, as in `scripts/summary.sh`. These were replaced by an independent port that matches.
- **Binary hash:** rebuild at `05a0ebd92` and `sha256sum` against `d05271011cd7…7771`.
- **Sweep checkers:** the crypto-side checkers, via `python3 tools/sheriff.py sweep --target crypto: --deep` with execution permitted.
- **CI after the PR:** `gh run list -R aburan28/crypto -w ic-autolab.yml`, plus the ic-leaderboard runs. GitHub access here is scoped to crypto-autoresearcher.
- **Re-renders:** `REPORT.typ`, `scripts/plots.py` and `scripts/build_ic_leaderboard.py --check` were not re-run (producer code).
- **Unseen curves:** whether `ic_frontier_campaign_20261015`, the source of r24/r28, informed the planner's rules informally (C14).

What this case does not establish

- That the planner chooses well outside the five one-change alternatives tested, or at sizes other than 23.5–29.6 bits. The same goes for binary, Koblitz or j = 0/1728 curves.
- That the counts reproduce on another host. No replay ran here, and the baseline arms were never replayed by anyone.
- Execution integrity. Hashes, seals and the producer's replays are not execution proofs (`docs/execution-proof-chain.md`).
- Any change of research status. That is the Coordinator's.

Verified delta, honestly sized

The study improves on no baseline: no `S` fell, and no leaderboard row moves (C11). Its value is calibration:

- At w = 0, the planner's rho outlook is within 1–18% (low) of the measured gap on six prime curves of 23.5–29.6 bits.
- At w = 0, the plan beats the five one-change alternatives tested.

Distance remaining to matched `rho.negation`: **12.6–25.8× at w = 0, and 15.6–30.7× at w = 1**, as lower-bound quotients. Rho remains cheaper on every curve.

Files

path	contents
`sweep.{json,md,svg}`	deterministic sweep; canary passed
`prior.md`	the prior
`findings.yaml`	the case; `check-case` passes
`readers/receipts.yaml`, `scripts/readers/receipts/`	validator: certificates, seals, plan.json, replays
`readers/stats.yaml`, `scripts/readers/stats/`	validator: blind statistics re-derivation, number trace
`readers/costmodel.yaml`, `scripts/readers/costmodel/`	red team: cost model, planner, class, leaderboard
`checks/session-checks.txt`	session's independence, chronology, copies and PDF-currency checks
`checks/REPORT.pdf.txt`	text extracted from the producer's PDF
`checks/workflow-ic-autolab.md`	session's workflow analysis
`checks/band-spotcheck.txt`, `scripts/session/band_spotcheck.py`	session's re-check of the w = 0/0.1/1 rho gap and the double-base/plan band
`problems/ecbench-replay-sampler-aliases-multi-arm-sessions.md`	problem record for F5 (all 72 producer replays fell on `plan-aa` or `rho-neg`)
`verdicts.svg`, `scripts/session/verdict_chart.py`	verdict chart
`report.typ`, `report.pdf`	this report rendered, with `sweep.svg` embedded